Resolution Geometry: The Conditional Naturality of General Relativity

A Unique Fixed Point of Declared Principles, with Each Declaration Tested Against Data

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Abstract

We replace the question "did nature choose general relativity?", which has no proof type, by one that does: is GR the *unique* law compatible with a short list of declared principles, and how tightly does data close the complement of each declaration? The answer has two parts.

The structural spine.

- The linear Einstein operator is selected by conservation, descent and second order.
- A complete on-shell ansatz finds exactly one gauge-invariant cubic vertex of a massless spin-2 field at 2, 4 and 6 derivatives each, and none otherwise. The 2-derivative vertex is Einstein–Hilbert. The 4-derivative (Gauss–Bonnet) vertex vanishes identically in four dimensions.
- All-order uniqueness at two derivatives is imported from Wald and from Boulanger, Damour, Gualtieri and Henneaux.

Hence, within the class of single massless spin-2, local, Lorentz-invariant, parity-even theories with at most two derivatives in $D = 4$, GR+Λ is the unique consistent theory.

The data. We test each declaration with light, real data.

- *Pantheon+.* From 1580 supernovae with the full statistical and systematic covariance, Λ is measured as $\Omega_m = 0.333$ (68%: $[0.315, 0.351]$). $\Lambda = 0$ is rejected by $\Delta\chi^2 = 48$ even with free curvature, and $w = -0.90^{+0.14}_{-0.16}$ is consistent with -1.
- *Binary pulsars.* Peters–Mathews applied to published timing parameters bounds any extra radiative channel beyond the two transverse-traceless modes below 1.6×10^{-4} of the quadrupole loss in the double pulsar.
- *GW170817.* The tensor speed equals c within $[-3.1 \times 10^{-15}, +6.5 \times 10^{-16}]$.
- *Cassini.* Brans–Dicke models with $\omega < 4 \times 10^4$ are excluded.

Two declarations remain untested by these data. The flat DGP braneworld (a fifth dimension) fits the supernovae as well as ΛCDM ($\Delta\chi^2 = -0.44$), so $D = 4$ is not tested at this level. Nor is the derivative-order cap that excludes R^3.

Conclusion. GR is not "given from above". It is the unique fixed point of named principles, each one an attackable hypothesis whose complement data excludes in explicit parameter regions. It is not proved optimal, and no competitor theory is eliminated as a whole.

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1 The question that has a proof type

"Nature selects GR as the optimal law" cannot be proved from observations, for three reasons:

- finite records leave unobserved continuations open, which is the local-mutation canary;
- "optimal" needs a declared criterion;
- Resolution Geometry proves that classification selects no dynamics (L128), and that the source remainder does not fix gravity.

Definition 1.1 (Conditional naturality)**.** A law $\mathcal{L}$ is *conditionally natural* for a declaration set $\mathcal{D}$ if two things hold:

(i) $\mathcal{L}$ is the unique consistent law satisfying $\mathcal{D}$;

(ii) for each $d \in \mathcal{D}$, the complement of d is excluded by data in a stated parameter region, or is declared as UNTESTED.

Measurements exclude *parameter regions*, not theories. For example, Brans–Dicke with $\omega \to \infty$ approaches GR within any tolerance. The ledger therefore reports excluded regions and surviving limits, never "theory X is eliminated".

2 The structural spine

Proposition 2.1 (Linear order; Both Sides)**.** *Among local second-order operators on a symmetric tensor, identical conservation together with descent to the gauge quotient leaves exactly the linearized Einstein operator, up to scale.*

Theorem 2.2 (Cubic universality; computed)**.** *Consider a single massless spin-2 field with on-shell three-point kinematics: $k_i^2 = k_i \cdot k_j = 0$, and polarizations $\varepsilon_i = e_i e_i$ with $e_i \cdot k_i = e_i \cdot e_i = 0$. Take the complete parity-even, Bose-symmetric ansatz in the six independent invariants and impose linearized gauge invariance on every leg. The number of gauge-invariant vertices by derivative count n is:*

n	0	1	2	3	4	5	6	≥ 7
ansatz size	1	0	6	0	3	0	1	0
invariant vertices	0	0	**1**	0	**1**	0	**1**	0

The vertices are:

- $n = 2$*:* $(\mathrm{YM})^2$*, the Einstein–Hilbert cubic vertex, unique up to the coupling;*
- $n = 4$*:* $\mathrm{YM} \cdot F^3$*, Gauss–Bonnet;*
- $n = 6$*:* $(F^3)^2$*, i.e.* R^3*.*

On explicit complex four-dimensional spinor-helicity kinematics, over both degenerate branches and all eight helicity configurations, the $n = 4$ *vertex vanishes (maximum* 5.6×10^{-17}*), while the* $n = 2$ *and* $n = 6$ *vertices do not.*

Nontrivial local vertices modulo field redefinitions and total derivatives correspond to on-shell three-point structures (imported). Beyond six derivatives, additional momenta enter only as $k_i \cdot k_j = 0$, so they amount to field redefinitions. The off-shell consistency of the GR candidate is checked separately: 32/32 deformed gauge-covariance identities and 8/8 second-order Bianchi identities.

Imported Theorem 2.3 (All orders at two derivatives [8, 9, 10])**.** *At most two derivatives, the consistent deformations of a free massless spin-2 field are Einstein–Hilbert* $+\Lambda$*. Several gravitons decouple into independent copies.*

Corollary 2.4. *In the class* $\mathcal{D}_0$ *of single massless spin-2, local, Lorentz-invariant, parity-even theories with at most two derivatives in* $D = 4$*, GR*$+\Lambda$ *is the unique consistent theory.*

Remark 2.5. Minimal hidden state (H-min) does *not* exclude R^3, which adds no propagating mode. Its exclusion is the derivative-order declaration. Gauss–Bonnet is excluded by $D = 4$ itself.

3 The declarations, one by one, against data

3.1 Λ is allowed by uniqueness; its value is measured (Pantheon+)

We use the public Pantheon+ release [1, 2]: 1580 supernovae with $z_{\mathrm{HD}} > 0.01$, Cepheid calibrators excluded, and the full statistical-plus-systematic covariance. The absolute-magnitude offset is marginalized analytically.

Model	Best fit	χ^2 ($\Delta\chi^2$)
flat ΛCDM	$\Omega_m = 0.333$ $[0.315, 0.351]$, $\Omega_\Lambda = 0.667$	1386.4
$\Lambda = 0$, flat ($\Omega_m = 1$)	—	2079.7 (+693)
$\Lambda = 0$, curvature free	$\Omega_m \to 0$ (empty boundary)	1434.6 (+48.2)
flat wCDM	$\Omega_m = 0.290$, $w = -0.896$ $[-1.055, -0.755]$	1386.0 (−0.44)
flat DGP (self-accelerating)	$\Omega_m = 0.221$	1386.0 (−0.44)

Intervals are 68%, and $\Delta\chi^2$ is relative to flat ΛCDM. The fitted Ω_m and w agree with the published Pantheon+ supernova-only constraints. $\Lambda > 0$ is required even when curvature is free. A cosmological constant is *permitted* by the uniqueness theorem, and its value is a measurement, not a selection.

3.2 Two radiating modes only (binary pulsars)

The quadrupole (two transverse-traceless modes) orbital decay is given by Peters–Mathews [7]. Applied to the published parameters of the two pulsars:

- **B1913+16** [3] ($P_b = 0.322997448918\,\mathrm{d}$, $e = 0.6171340$, masses 1.438 and $1.390\,M_\odot$): the computation gives $\dot{P}_b^{\mathrm{GR}} = -2.4021 \times 10^{-12}$, against the published -2.40263×10^{-12}. With the intrinsic $-(2.398 \pm 0.004) \times 10^{-12}$, the ratio is 0.9983 ± 0.0017.
- **J0737−3039A/B** [4]: the computation gives -1.2478×10^{-12}, and the published ratio is 0.999963 ± 0.000063.

Any additional radiative channel, i.e. hidden radiating modes beyond the two TT modes, therefore carries less than 1.6×10^{-4} of the quadrupole loss at 2σ in the double pulsar. This bounds a parameter region of radiating extensions. It does not eliminate them all; for example, weakly coupled scalars with small sensitivity differences survive.

3.3 A single massless, luminal spin-2 field (GW170817)

Take the delay 1.74 s between GW170817 and GRB 170817A, the conservative distance 26 Mpc, and an intrinsic emission lag of 0–10 s [5]. Then

$$-3.09 \times 10^{-15} \leq c_T/c - 1 \leq 6.5 \times 10^{-16},$$

which reproduces the published bound. The speed-implied graviton mass bound is weak ($3 \times 10^{-20}\,\mathrm{eV}$ at 100 Hz); dispersion bounds are stronger. Theories whose tensor speed differs from c in the relevant regime are excluded in this region.

3.4 Structure channel $\gamma = 1$ (Cassini)

From $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [6], the 2σ lower limit gives Brans–Dicke $\omega > 3.9998 \times 10^4$. The limit $\omega \to \infty$ survives.

3.5 Untested by these data

- $D = 4$. The flat DGP braneworld [11] fits Pantheon+ as well as ΛCDM, so supernovae alone cannot distinguish them: a correspondence at this resolution. Other probes (expansion-history combinations, growth of structure) would be needed. They are not executed here.
- **At most two derivatives.** The R^3 coefficient is not constrained by any data used here.

4 Main theorem and ledger

Main Theorem (Conditional naturality of GR). *GR+Λ is conditionally natural for the declaration set $\mathcal{D}_0$:*

(i) it is the unique consistent theory in $\mathcal{D}_0$: linear and cubic orders computed, all orders imported;

(ii) the declarations are constrained by data as in the ledger below.

Declaration	Complement	Data result	Status
massless, luminal spin-2	$c_T \neq c$	$\lvert c_T/c - 1\rvert \lesssim 3 \times 10^{-15}$	constrained
two TT radiating modes	extra radiative channels	$< 1.6 \times 10^{-4}$ of quadrupole loss	constrained
$\gamma = 1$ (structure)	scalar admixture	BD $\omega > 4 \times 10^4$	constrained
Λ (allowed)	$\Lambda = 0$	$\Delta\chi^2 = 48$ with free curvature	measured
$w = -1$	dynamical dark energy	$w \in [-1.055, -0.755]$ (SN only)	consistent
$D = 4$	DGP leakage	$\Delta\chi^2 = -0.44$	UNTESTED
at most 2 derivatives	R^3	—	UNTESTED
locality, Lorentz, parity	—	partly via c_T	declared

No-Go 4.1 (What is not claimed).

- That nature *optimizes* anything: there is no criterion-free optimality.
- That any competitor theory is eliminated as a whole. Only regions are excluded; $\omega \to \infty$, tiny masses and screened sectors survive.
- That $D = 4$ or the derivative-order cap is tested.
- A defeat of realism.
- A complete cosmological analysis: there is no BAO, CMB or growth data, and H_0 is not used.

5 What this means for Resolution Geometry

- The question "is GR given from above?" is replaced by a ledger. Each declaration is named, and each is either measured, constrained, or explicitly UNTESTED.
- What remains for "from above" is the UNTESTED rows, namely dimension and derivative order, together with locality, Lorentz invariance and parity. All of them are hypotheses that experiments can attack.
- Resolution Geometry contributes the typing and the structural computations. It does not derive the law. The print's no-go stands: the field equation is selected, and the selection is now shown to be unique and empirically pinned.

6 Reproduction and limits

Reproduction.

- `rg_cubic_universality.py` (sympy): the cubic ansatz and the 4D check.
- `rg_gr_naturality_data.py` (numpy, scipy): D1–D4.

Both give byte-identical reruns.

Data.

- The Pantheon+ files are downloaded from the public release; their SHA-256 hashes are recorded in the results.
- The pulsar, GW170817 and Cassini inputs are published values, cited.

Limits.

- The supernova fits use profile likelihoods on a grid, not MCMC.
- The pulsar mass inputs are rounded, which leaves a 0.02% gap to the published GR $\dot{P}_b$.
- The on-shell equivalence and the all-orders uniqueness are imported.

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